

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: F.Y.B.Tech.

Branch :Common to all Branches

Semester : I

Subject Code & Name: BTES103G Engineering Graphics

Max Marks: 60

Date:11/02/2025

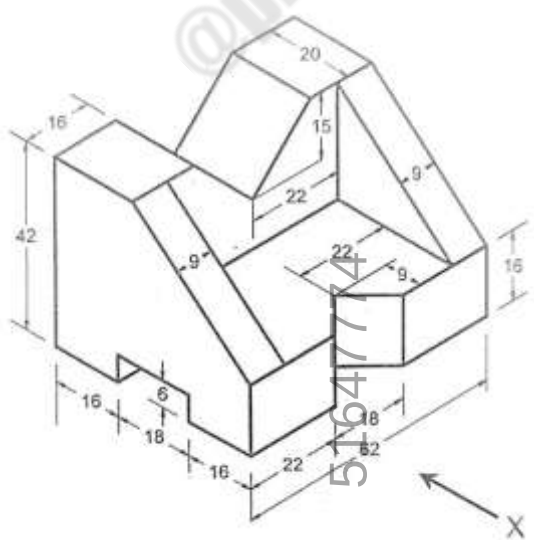
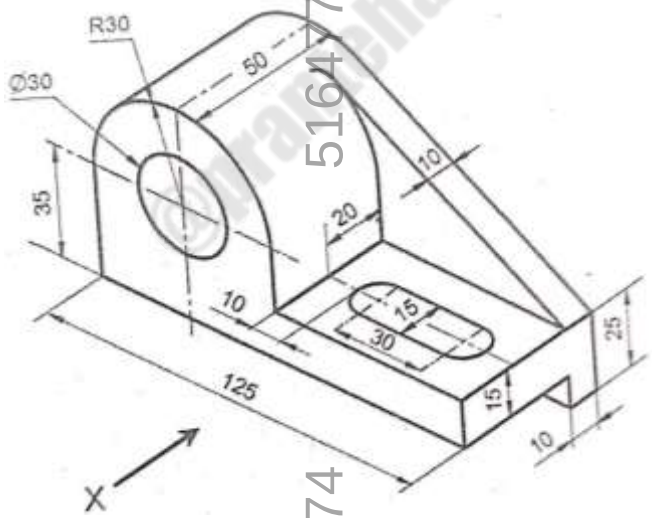
Duration: 4 Hr.

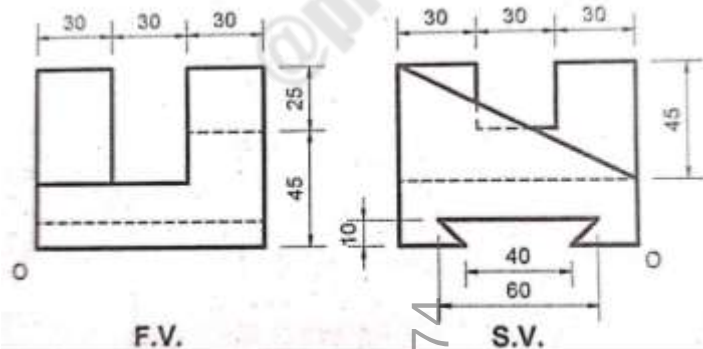
Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The polygon having eleven sides is called ----- a). Nonagon b) Decagon c) Un decagon d) Dodecagon	CO-I	1
2	The following line is used for dimension line a) Continuous thick b) Continuous thin c) Chain thin line d) Short zigzag thin line	CO-I	1
3	If object is in third quadrant it will be ---- a) Below HP & behind VP b) In front of VP & above HP c) Above HP & behind VP d) Below HP & In front of VP	CO-II	1
4	If point P is , in front of VP & below HP , is in ----- a) First quadrant b) Second quadrant c) Third quadrant d) Fourth quadrant	CO-II	1
5	If projectors are parallel to each other & perpendicular to reference plane is called a) Orthographic projection b) Isometric projection c) Oblique projection d) Perspective projection	CO-III	1
6	In Orthographic projection , Side view is projected on a) vertical plane b) Horizontal plane c) Profile plane d) Auxiliary plane	CO-III	1
7	When a point is above HP & in front of VP, the point is in Which quadrant ?	CO-II	1

	a.First	b. Second	c. Third	d. Fourth		
8	The length of F.V. and T.V. of the line, when line is inclined to both the plane are ----than the true length. (T.L.) of the line.				CO-III	1
	a) Less	b) Greater	c) Equal	d) Two times greater		
9	If circular lamina is equally made inclined to both principal planes , its plan and elevation seen as -----				CO-IV	1
	a) Circle	b) Square	c) Ellipse	d) Rhombus		
10	Name the regular polyhedron that is formed with four equilateral triangles.				Co-IV	1
	a) Tetrahedron	b) Triangular pyramid	c) Dodecahedron	d) Icosahedrons		
11	The trace of a section plane is				CO-V	1
	a) a curve	b) a straight line	c) a circle	d) a rectangle		
12	The Rectangular axes are inclined at ----- degree to each other.				CO-V	1
	a) 60	b) 90	c) 120	d) 150		
Q. 2	Solve the following.					12
A)	Draw the regular pentagon of 30 mm side, use any accurate method.				CO-I	6
B)	What is dimensioning? explain methods of dimensioning with neat sketch.				CO-I	6
Q.3	Solve, any one of the following.					12
A)	Draw the front view in X- direction, Top view and side view of the following sketch by angle Method of Projection.				CO-II	12

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B)	OR Figure shows, an isometric view, looking in X direction, draw 1. Elevation 2. Plan 3. Visible End View 	CO-II	12
Q. 4	Solve Any Two of the following.		12
A)	A straight line , 90 mm long , one end ,is 20 mm above HP and 25 mm in front of VP. The line is inclined at 25 degree to HP and 40 degree to VP. Draw projection .	CO-III	6
B)	A hexagonal lamina , 30 mm side , one corner is resting in VP and the diagonal is passing through , resting corner is parallel to HP and 45 degrees inclined to VP. Draw three views.	CO-III	6

C)	A pentagonal plate , 30 mm side , one side is in HRP . The surface of plate is made inclined by 45 degrees to HRP and perpendicular to VP draw three views.	CO-III	6
Q.5	Solve Any Two of the following.		12
A)	A hexagonal pyramid , base 30 mm side and length of axis , 60 mm , one corner of the base is in HP. The axis of pyramid is parallel to VP and 45 degrees inclined to HP. Draw Projection.	CO-IV	6
B)	A right circular cone , base 50 mm diameter and length of axis 70 mm, the axis of cone is inclined by 45 degrees to HRP and parallel to FRP. Draw projection.	CO-IV	6
C)	A square pyramid , base 30 mm side and 60 mm height , all base edges are equally inclined with FRP. It is cut by A.I.P., inclined at 45 degree to HRP and perpendicular to FRP. The cutting plane is bisecting the axis. Draw sectional top view and true shape of a section.	CO-V	6
Q. 6	Solve Any One of the following.		12
A)	Figure shows , two orthographic views of an object , take O as a origin and draw isometric view. 	CO-V	12
B)	Figure shows , two orthographic views of an object , take O as a origin and draw isometric view.	CO-V	12

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	*** End ***	

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